The Cartesian Equation of a Plane in R^3

Theorem: If $\vec{u}$ and $\vec{v}$ are not collinear vectors and $\vec{n}$ is a vector perpendicular to both $\vec{u}$ and $\vec{v}$, then $\vec{n}$ is perpendicular to EVERY vector coplanar with $\vec{u}$ and $\vec{v}$.

Proof:

Normal Vector – A vector that is _____ to every vector in a plane is called a normal vector to the plane.

Let $\vec{u}$ and $\vec{v}$ be two non-collinear vectors in the plane.

Then the normal $\vec{n}$ =

Let $P_0(x_0, y_0, z_0)$ be a known point on the plane.
Let $P(x, y, z)$ be any point on the plane.

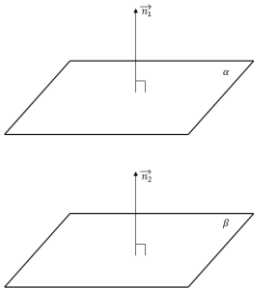
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Example 1: Determine the Cartesian equation of the plane
$\vec{r} = (2,-1,4) + s(1,-3,0) + t(-4,1,1),\ s,t \in \Re$

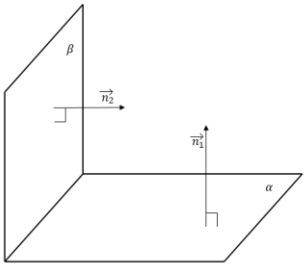
Example 2: Determine the Cartesian equation of the plane through the points $A(2,3,-1)$, $B(-1,2,2)$ and $C(3,-1,2)$

Note: Normals are used to determine the angle between two planes.

a) Parallel Planes



b) Perpendicular Planes



c) Intersecting Planes

